Variations in solution chemistry during calcium-deficient and stoichiometric hydroxyapatite formation from $CaHPO_4 \cdot 2H_2O$ and $Ca_4(PO_4)_2O$

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This study explores the mechanistic paths taken when calcium-deficient hydroxyapatite, CDHAp (Ca/P = 1.50), and stoichiometric hydroxyapatite, SHAp (Ca/P = 1.67), form by reaction between particulate calcium phosphate salts. The acidic reactant was $CaHPO_4 \cdot 2H_2O$ (DCPD) and the basic reactant was $Ca_4(PO_4)_2O$ (TetCP). Variations in pH, calcium and phosphate concentrations, and the solids present during apatite formation, were determined as functions of reaction temperature (25.0°, 37.4°, and 50.0°C) and time. It was found that the dissolution of TetCP was rate limiting for both hydroxyapatite (HAp) compositions at all three temperatures. However, the retrograde solubility and incongruent dissolution of DCPD became increasingly important in influencing the kinetics as the reaction temperature was increased. An amorphous intermediate phase was observed regardless of the HAp stoichiometry. The solutions from which the SHAp formed approached equilibrium at much shorter reaction times (1–2 days) than those from which the CDHAp formed. The latter continued to display changes in pH and in calcium and phosphate concentrations for 6 months. CDHAp was shown to be a thermodynamically stable phase. The dissolution of CDHAp is incongruent, showing a Ca/P molar ratio in solution less than 0.5. © 1997 John Wiley & Sons, Inc.

INTRODUCTION

Calcium phosphates, specifically apatites, are suitable for a number of biomedical applications involving hard tissue replacement, augmentation, and defect filling. From the standpoint of biological compatibility and minimal immune response, apatites are ideal candidates as biomaterials due to their similarities in composition to native hard tissues. The mineral phase of hard tissue is formed of apatites that contain varying amounts of CO_3^{2-}, F^-, Mg^{2+}, Sr^{2+}, K^+, and Na^+;[1] the amount of each ion depends on the species, age, and type of hard tissue as well as on the region sampled.[2] HAp exists over a range of compositions typically characterized by the Ca/P molar ratio. For unsubstituted apatites (the CaO-P_2O_5-H_2O system) this is believed to range between 1.50 and 1.67.[3,4] However, Ca/P molar ratios down to 1.33 have been reported.[5,6] The lowest attainable limit of Ca/P in HAp may be a function of the specific synthesis route employed. Therefore, the composition of synthetic apatite, such as carbonate content and Ca/P ratio, can be varied to suit specific *in vivo* applications. This permits tailoring of properties, such as pore structure and resorption rate.

A relatively recent development that also makes synthetic apatites desirable candidates as replacements for hard tissue is the discovery that they can be formed at physiological temperatures[7] by reactions between mixtures of particulate DCPD ($CaHPO_4 \cdot 2H_2O$) or DCP ($CaHPO_4$) and TetCP [$Ca_4(PO_4)_2O$]. Further work has demonstrated formulations could set and reach complete reaction within clinically relevant periods of time.[8–10] These reactants can be constituted as a formable paste that hardens into a monolithic form as reaction proceeds.

The present investigation analyzes the formation of SHAp [$Ca_{10}(PO_4)_6(OH)_2$] and CDHAp [$Ca_9HPO_4(PO_4)_5OH$] by reactions between DCPD and TetCP (see Eq. 1 and 2) at various temperatures spanning physiological variations. While *in vivo* formation of HAp by such reactions may be complicated by the presence of the constituents of biological fluids, formation in water provides insight into the mechanistic reaction path taken. This, in turn, facilitates tailoring these reactions for *in vivo* use.

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MATERIALS AND METHODS

Precursor synthesis

TetCP was synthesized from reagent grade DCP (Fisher Scientific, Pittsburgh, PA) and precipitated calcium carbonate ($CaCO_3$, Fisher Scientific, Pittsburgh, PA) by a high temperature solid state reaction.[9] DCPD was precipitated via an acid-base reaction between calcium hydroxide and monocalcium phosphate monohydrate [MCPM, $Ca(H_2PO_4)_2 \cdot H_2O$].[9] DCPD and TetCP were combined in 2:1 and 1:1 molar ratios and milled together to constitute HAp precursor powders with Ca/P molar ratios of 1.50 and 1.67. Reactions shown in Equations 1 and 2 show the formation of CDHAp (Ca/P = 1.50) and SHAp (Ca/P = 1.67), respectively.

$$6CaHPO_4 \cdot 2H_2O + 3Ca_4(PO_4)_2O \xrightarrow{H_2O} 2Ca_9(HPO_4)(PO_4)_5OH + 13H_2O \quad \text{(Eq. 1)}$$

$$2CaHPO_4 \cdot 2H_2O + 2Ca_4(PO_4)_2O \xrightarrow{H_2O} Ca_{10}(PO_4)_6(OH)_2 + 4H_2O \quad \text{(Eq. 2)}$$

Solution chemistry

Analyses of the variations in the solution chemistry during HAp synthesis according to Equations 1 and 2 as a function of time were performed at 25.0°, 37.4°, and 50.0°C. In a typical experiment a 400 mL jacketed reaction vessel was filled with 350 mL of deionized water. The temperature was controlled to ±0.1°C. Upon thermal equilibration 70.0 g of HAp precursor powder was added to the water under constant stirring. N_2 was bubbled through the suspension throughout the reaction to minimize the presence of CO_2. The pH was measured throughout the reaction using a combination glass electrode (Orion Inc., Boston, MA) calibrated using buffer solutions of pH 4.0 and 10.0. A datum point was collected every 20 sec over the first 24 h of reaction using an Orion 920 pH meter interfaced with a computer. Reactions at 50°C were run for 48 h. After 24 h of reaction at 25.0° and 37.4°C the slurries were sealed in Nalgene® bottles and maintained at 25.0° and 37.4°C. The pH values of these samples periodically were measured over a 6 month period. Suspension aliquots were removed at selected times and filtered through 0.22 μm membranes (Millipore, GV type, Millford, MA). Solutions were analyzed for calcium and phosphate using DC plasma emission spectroscopy. The solids immediately were frozen in liquid nitrogen and freeze-dried for X-ray diffraction studies.

Solids analysis

Solids were analyzed using a powder diffractometer (Scintag, Inc., Sunnyvale, CA) with copper kα radiation to determine the solids present as a function of time. Powdered samples were packed into zero-background quartz slides and scanned over a 2θ range of 10°–35° using a step size of 0.02° 2θ. Selected samples also were scanned over a 2θ range of 3°–10° to check for the presence of OCP [$Ca_8(HPO_4)_2(PO_4)_4 \cdot 5H_2O$]. Samples equilibrated for 4 weeks (25.0° and 37.4°C) and 48 h (50.0°C) were analyzed for carbonate by carbon coulombetry.

RESULTS AND DISCUSSION

Variations in pH

This section discusses the variations in pH during HAp formation and the related changes in heat evolution[9] and solids present. Figure 1 plots the short- and long-term variations in pH during the formation of SHAp and CDHAp by Equations 1 and 2 at 25°, 37.4°, and 50°C. All reactions occur in stages, characterized by changes in pH, that vary in magnitude and duration depending on the stoichiometry and temperature. These stages are most clearly observable for HAp formation at 25°C due to the slower kinetics; discussion of Figure 1 will emphasize HAp formation at this temperature.

At 25°C a pH value near 7.7 is reached within the

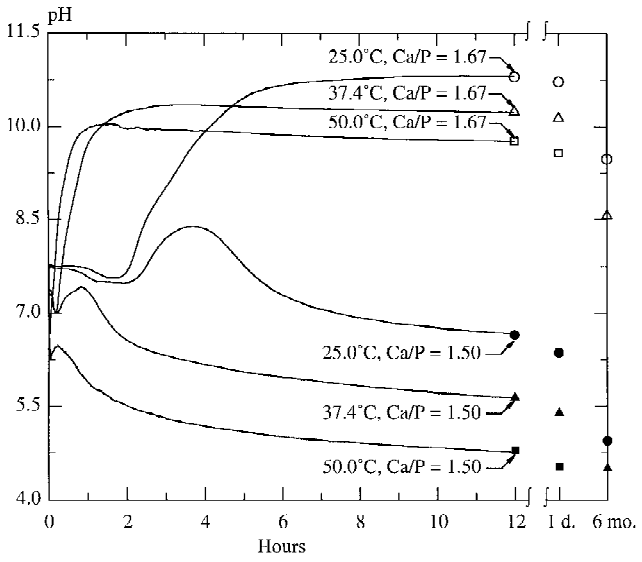


Figure 1. pH as a function of time when HAp having Ca/P molar ratios of 1.50 or 1.67 forms at 25.0°, 37.4°, or 50.0°C. The curves are labeled with their respective bulk Ca/P molar ratio and reaction temperatures.

first few minutes after mixing for both bulk compositions. However, the stoichiometric composition reaches a slightly higher value due to the presence of a higher proportion of the basic reactant, TetCP. The pH remains nominally constant over the next ~0.5 and ~1.0 h during the formation of CDHAp and SHAp, respectively. During this time isothermal calorimetric studies[9] have shown an induction period in heat evolution, as illustrated in Figure 2, which plots heat evolution in Watts/mol of HAp formed versus time. During this short induction period small amounts of heat are evolved for both stoichiometries. Differences between the unreacted precursor powders and those reacting for these lengths of time cannot be detected by X-ray diffraction [see Fig. 3(a) and 4(a)].

After these induction periods, there are decreases in pH indicating preferential dissolution of the acidic reactant. The stoichiometric composition decreases to a pH value of ~7.55 at 1.5 h while the calcium-deficient composition attains a slightly lower pH of ~7.50 by 1.2 h. During this period of near steady-state pH (7.55 for SHAp between 1.5 and 1.75 h and 7.50 for CDHAp between 1.2 and 2.0 h) progress of reaction becomes evident by X-ray diffraction analyses [see Fig. 3(b) and 4(b)]. The proportions of reactants are observed to decrease as indicated by the reduction in their diffraction intensities. The first signs of reaction products also occur during this time. Indications of both a small crystallite HAp and a noncrystalline calcium phosphate (NCP) can be seen in Figures 3(b) and 3(c) and

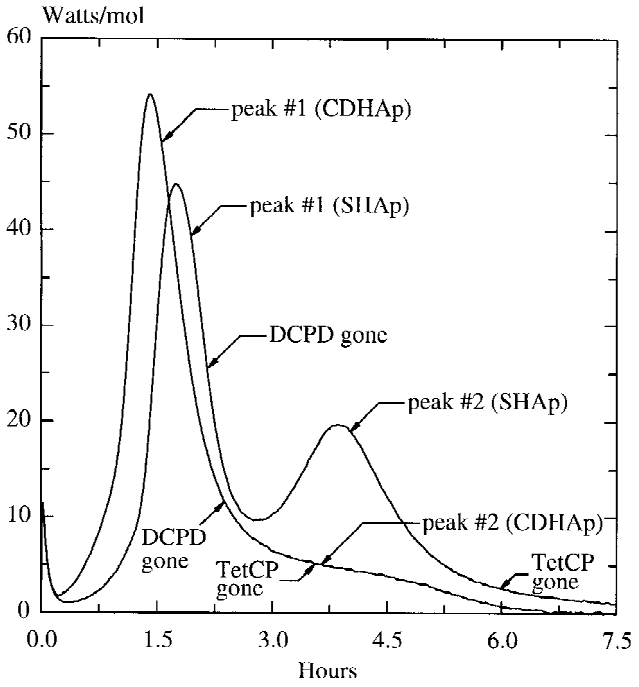


Figure 2. Heat evolution (dQ) in Watts/mol of HAp formed as a function of time for the formation of both CDHAp and SHAp at 25.0°C. Significant events that occur during the course of the reaction are highlighted.

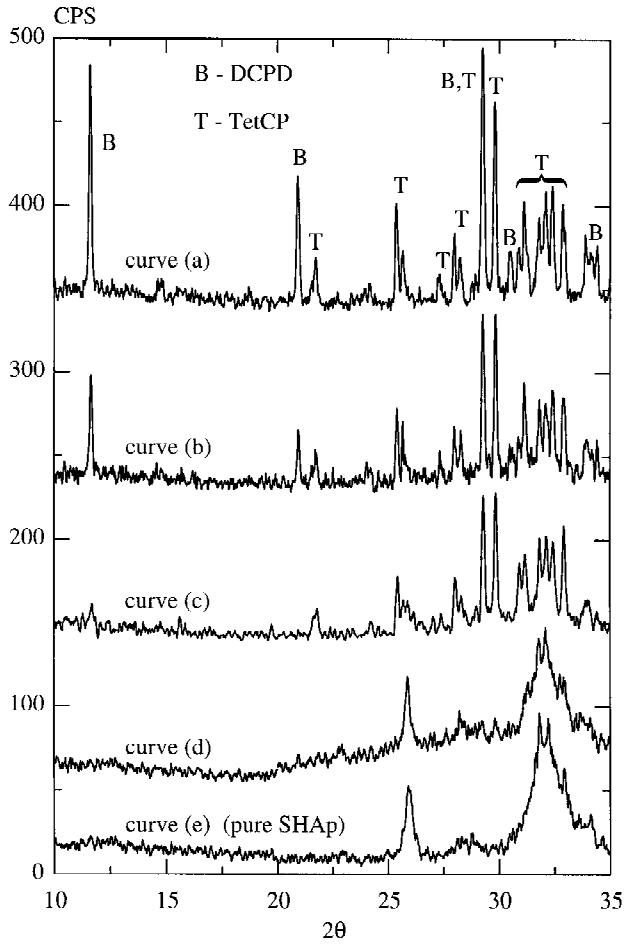


Figure 3. X-ray diffraction patterns as a function of time during the formation of SHAp at 25.0°C: (a) 1.5 h (b) 1.75 h (c) 2.1 h (d) 6.0 h (e) 24 h.

in Figures 4(b) and 4(c) associated with a broad amorphous background. Consistent with the initial decrease in pH, X-ray diffraction also indicates that the DCPD is preferentially consumed.

Eventually the pH values begin to increase, occurring after ~1.75 and 2.0 h for the SHAp and CDHAp compositions, respectively. Small proportions of DCPD still are present when this occurs. X-ray diffraction indicates the complete consumption of DCPD at ~2.1 and ~2.4 h for the SHAp and CDHAp compositions, respectively [see Fig. 3(c) and 4(c)]. This indicates the dissolution of TetCP is rate limiting. For both bulk stoichiometries the reaction consuming all of the DCPD and some of the TetCP is associated with the first calorimetric reaction peak (labeled peak #1 in Fig. 2). This reaction has been qualitatively expressed for the two bulk compositions (Ca/P = 1.50 and 1.67) by Equations 3 and 4, respectively.[9] It has been shown by others[11] that the Ca/P ratio of unwashed amorphous calcium phosphate precipitates is variable and changes with pH. At pH values below 7.5 the Ca/P ratio should be less than 1.35.[11] Therefore, the Ca/P

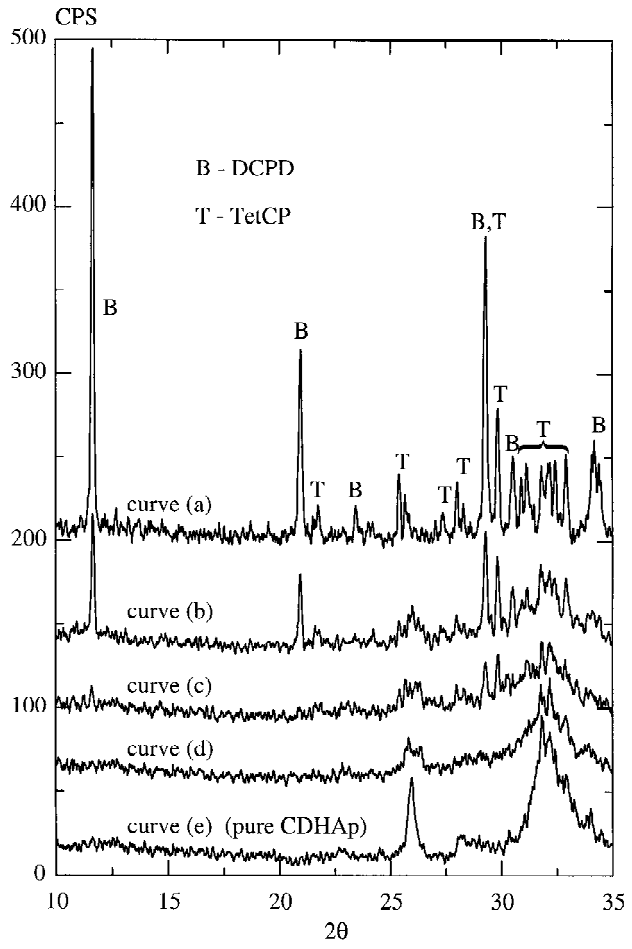


Figure 4. X-ray diffraction patterns as a function of time during the formation of CDHAp at 25.0°C: (a) 1.2 h (b) 2.0 h (c) 2.4 h (d) 3.50 h (e) 24 h.

ratio of the NCP intermediate formed by these reactions has an upper limit of 1.35. The preferential consumption of DCPD combined with the pH values observed during this reaction stage indicates the NCP phase that forms should have a Ca/P < 1.35 and not 1.5, as we previously proposed.[9]

$$6CaHPO_4 \cdot 2H_2O + (3-x)Ca_4(PO_4)_2O \rightarrow \text{NCP } (1.2 \leqslant Ca/P \leqslant 1.35) + \text{HAp } (1.50 \leqslant Ca/P \leqslant 1.67) \quad \text{(Eq. 3)}$$

$$2CaHPO_4 \cdot 2H_2O + (2-x)Ca_4(PO_4)_2O \rightarrow \text{NCP } (1.2 \leqslant Ca/P \leqslant 1.35) + \text{HAp } (1.50 \leqslant Ca/P \leqslant 1.67) \quad \text{(Eq. 4)}$$

The increases in pH just prior to the complete consumption of DCPD are due to the dissolution of TetCP, and a continuous decrease in its diffraction intensities occurs as it reacts with the intermediate products formed in Equations 3 and 4. These reactions[9] are shown by Equations 5 (Ca/P = 1.50) and 6 (Ca/P = 1.67) and are associated with the second calorimetric peaks shown in Figure 2 (labeled peak #2). The greater apparent area under peak #2 for the SHAp reaction is related to the presence of a higher proportion of the rate limiting reactant, TetCP.[9]

$$x Ca_4(PO_4)_2O + \text{NCP } (1.2 \leqslant Ca/P \leqslant 1.35) + \text{HAp } (1.50 \leqslant Ca/P \leqslant 1.67) \rightarrow Ca_9(HPO_4)(PO_4)_5OH \quad \text{(Eq. 5)}$$

$$x Ca_4(PO_4)_2O + \text{NCP } (1.2 \leqslant Ca/P \leqslant 1.35) + \text{HAp } (1.50 \leqslant Ca/P \leqslant 1.67) \rightarrow Ca_{10}(PO_4)_6(OH)_2 \quad \text{(Eq. 6)}$$

X-ray diffraction analysis during CDHAp formation indicates TetCP is consumed by ~3.5 h [see Fig. 4(d)]. This is coincident with the time at which the pH reaches its maximum value in Figure 1. However, TetCP is not consumed until ~6 h when SHAp is forming [see Fig. 3(d)]. The greater proportion of TetCP in the SHAp mixture coupled with its slower reactivity accounts for its being observed for a longer period than when CDHAp forms. Again as TetCP consumption is complete, the pH approaches its maximum value. X-ray diffraction also suggests the persistence of NCP after the disappearance of both DCPD and TetCP [see Fig. 3(d) and 4(d)]. This is consistent with the calorimetric curves in Figure 2, which show extended periods of slow reaction after completion consumption of the reactants is observed by X-ray diffraction. The quantity of heat evolved between the time of disappearance of TetCP and complete reaction by isothermal calorimetry is slightly higher for the formation of CDHAp. Assuming this heat is due to the crystallization of the amorphous intermediate suggests that a greater quantity of NCP is formed at the lower Ca/P composition. Within several hours after the precursors disappear, the amorphous background decreases, indicating conversion to an apatitic product [see Fig. 3(e) and 4(e)].

A major difference between the SHAp and CDHAp compositions is the change in pH after TetCP consumption (Fig. 1). As Figure 2 shows, the pH values of solutions equilibrating with SHAp at 25.0° and 37.4°C continually decrease over extended equilibration times at both temperatures. This may be attributed to the slow incorporation of small amounts of carbonate. Carbon coulombetry results indicate the presence of 0.42, 0.48, and 0.53 weight percent carbonate in SHAp samples equilibrated as follows: 25.0° (4 weeks), 37.4° (4 weeks), and 50.0°C (2 days). The pH values of solutions in contact with the CDHAp samples also decreased over extended periods of time (Fig. 1). However, the continuous drop in pH in these samples cannot be attributed to atmospheric carbon dioxide as in the stoichiometric samples. The uptake of carbonate is negligible at these pH values, as supported by carbon coulombetry. These findings indicate the solution and the CDHAp are not in equilibrium and that long periods of equilibration are required. This results from the incongruent dissolution of CDHAp and subsequently will be discussed in greater detail.

Variations in calcium and phosphate concentrations

The calcium and phosphate concentrations were analyzed during and subsequent to the formation of SHAp and CDHAp at 25.0°, 37.4°, and 50.0°C to allow an assessment of their solubility behavior. Although occurring at shorter times with increasing temperature, the variations in calcium and phosphate during HAp formation were similar at each stoichiometry regardless of the temperature (25.0°, 37.4°, or 50.0°C). Therefore, a detailed discussion will be limited to 25.0°C.

Stoichiometric HAp

Figure 5 plots the calcium and phosphate concentrations during the formation of SHAp at 25.0°C. The variations observed coincide with those in pH and in solids present, as previously discussed. Although the pH remains nominally constant during the first hour, the calcium and phosphate concentrations decrease. It is during this time that the solution becomes critically supersaturated with respect to HAp and nucleation initiates. After 1 h, the pH begins to decrease due to preferential DCPD dissolution, and the calcium and phosphate concentrations increase. Isothermal calorimetry[9] indicates the onset of the first reaction peak at this time (see Fig. 2). Coincidentally, after about 1.75 h the calcium and phosphate concentrations reach their maxima, the pH reaches a minimum, the rate of heat evolution during the first reaction peak reaches a maximum, and X-ray diffraction indicates the presence of a noncrystalline calcium phosphate phase. Elevated ion concentrations favor the formation of an amorphous calcium phosphate, indicating that its precipitation will be most favored during this period of the reaction.

Shortly before X-ray diffraction analysis shows the disappearance of the remaining DCPD (~2.1 h), the pH begins to rise (~1.75 h) and the calcium and phosphate concentrations start to decrease. This continues until about 5 h of reaction. Between 5 and 6 h the calcium concentration increases slightly and then slowly decreases until it reaches a minimum value of ~0.25 m*M* after ~24 h. Longer reaction times do not result in significant changes in the measured ion concentrations (Fig. 5). Although X-ray diffraction indicates both precursors have been consumed by 5–6 h, changes in the calcium concentration between 5 and 24 h are consistent with the conversion of an NCP intermediate phase to HAp. The amorphous phase present at 6 h [Fig. 3(d)] can be observed at 12 h but not at 24 h [Fig. 3(e)].

Reaction at 37.4°C (Fig. 6) exhibits changes in ion concentrations similar to those observed at 25.0°C; however, these variations occur more rapidly. X-ray diffraction indicates DCPD and TetCP are consumed by 0.35 and 1.5 h, respectively. Reaction at 50.0°C (Fig.

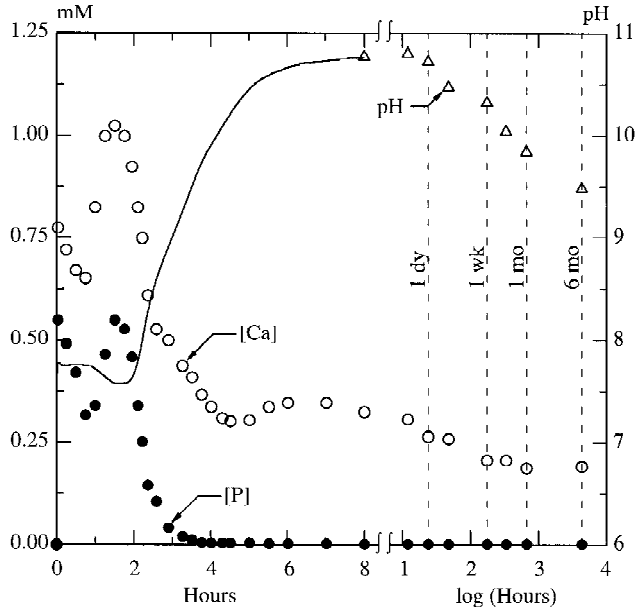


Figure 5. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of SHAp at 25.0°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in m*M* (10^{-3} moles/liter). The pH (solid line to 8 h and then open triangles) values are plotted on the y_2 axis.

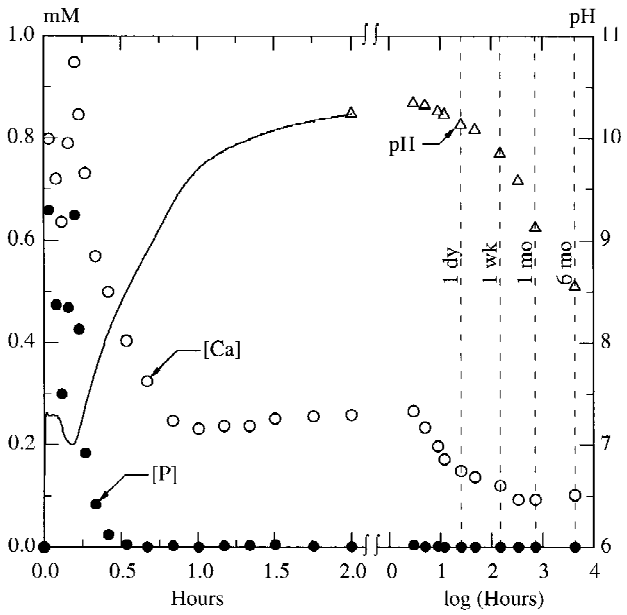


Figure 6. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of SHAp at 37.4°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in m*M* (10^{-3} moles/liter). The pH (solid line to 2 h and then open triangles) values are plotted on the y_2 axis.

7) occurred so rapidly the initial variations in calcium and phosphate concentrations were not observed. DCPD is consumed within 5 min and TetCP within 0.75 h.

Calcium-deficient HAp

Figure 8 shows the changes in calcium and phosphate concentrations during the formation of CDHAp at 25.0°C. The variations in these concentrations differ from those during the formation of SHAp. The calcium and phosphate concentrations increase during the first 1.5 h of reaction compared to an initial decrease when SHAp forms. Based on pH changes the opposite trend would be expected to occur. The difference is likely related to the larger proportion of DCPD present when CDHAp is to be formed. Although the calcium and phosphate concentrations increase during the initial formation of CDHAp while they decrease during SHAp formation, both occur at nomimally constant pH values (Fig. 1). Maxima in calcium and phosphate concentrations occur at about 1.25 h; this is approximately the time of occurrence of the maximum rate of heat output during the first reaction peak (Fig. 2).[9] After reaching these maxima, the calcium and phosphate concentrations decrease over the next 2.75 h. This coincides with the increase in pH and with the continuous precipitation of HAp/NCP. X-ray diffraction analysis indicates the presence of HAp and NCP by 2.0 h. DCPD and TetCP are consumed at ~2.4 h and 3.5–4.0 h, respectively. Upon

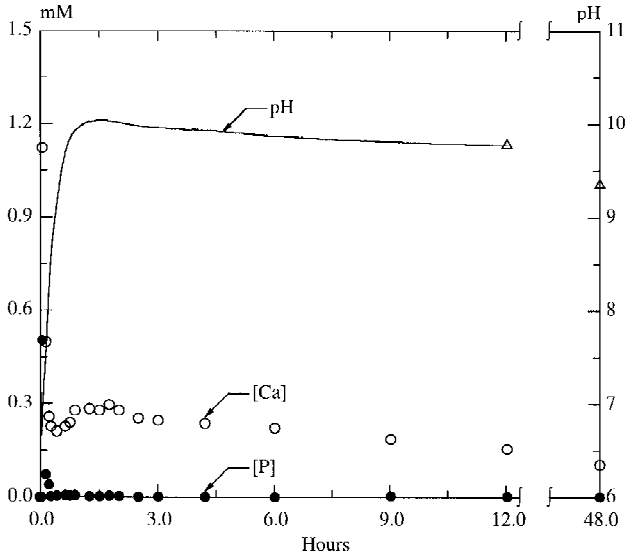


Figure 7. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of SHAp at 50.0°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in mM (10^{-3} moles/liter). The pH (solid line to 12 h and then open triangles) values are plotted on the y_2 axis.

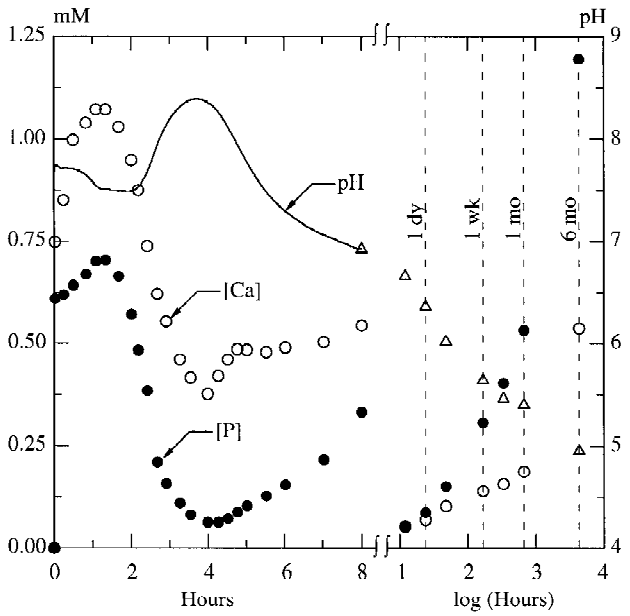


Figure 8. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of CDHAp at 25.0°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in mM (10^{-3} moles/liter). Calcium and phosphate concentrations plotted after the break in time (∫∫) are $\frac{1}{10}^{th}$ of their actual value (for example, P at 1 week is actually 3.1 mM and not 0.31 mM). The pH (solid line to 8 h and then open triangles) values are plotted on the y_2 axis.

TetCP consumption, the calcium and phosphate concentrations reach their minimum values and the pH reaches its maximum value. X-ray diffraction indicates NCP is present along with small crystallites of HAp. For 2.5–3.0 h after X-ray diffraction has indicated the consumption of both precursor phases, isothermal calorimetry indicates continued heat evolution due to the conversion of NCP to CDHAp.

X-ray diffraction indicates that the CDHAp crystallites mature and grow during the first 24 h [Fig. 4(d, e)]. Subsequent to the consumption of TetCP, there is a continuous increase in the calcium and phosphate concentrations along with a decrease in pH. As indicated in Figure 8, the pH and the calcium and phosphate concentrations significantly change over a period of 6 months. Thus, while SHAp approaches equilibrium with its solution within days, the equilibration of CDHAp is not achieved within 6 months.

CDHAp formation at 37.4° (Fig. 9) and 50.0°C (Fig. 10) follows the same mechanistic path as at 25°C, but, again, events occur at shorter times. At 37.4°C the disappearance of the reactant phases, changes in pH and in calcium and phosphate concentrations (Fig. 9), and the reaction peaks in the calorimetric curve[9] all coincide. X-ray diffraction indicates DCPD and TetCP are consumed by ~0.6 and ~0.9 h, respectively. As observed during the formation of SHAp, reactions to form CDHAp at 50.0°C occurred so rapidly that initial

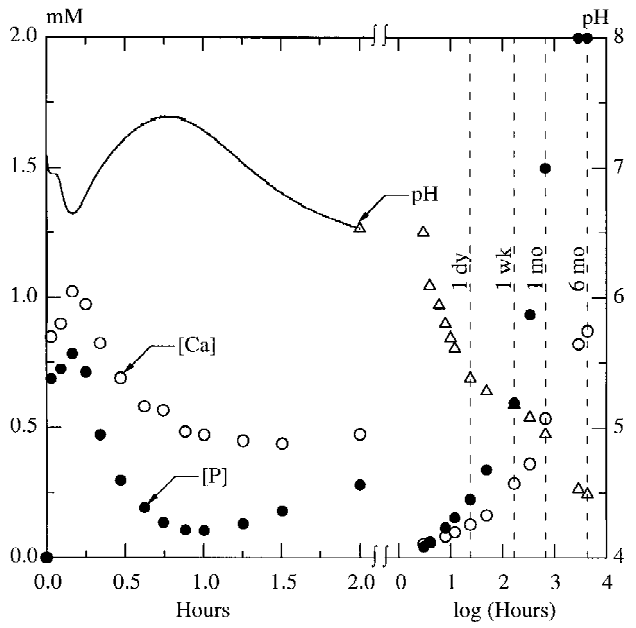


Figure 9. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of CDHAp at 37.4°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in m*M* (10^{-3} moles/liter). Calcium and phosphate concentrations plotted after the break in time (∫∫) are $\frac{1}{10}^{th}$ of their actual value (for example, P at 6 months is actually 20 m*M* and not 2.0 m*M*). The pH (solid line to 2 h and then open triangles) values are plotted on the y_2 axis.

variations in the solution observed at the lower temperatures could not be discerned (Fig. 10). X-ray diffraction indicates that DCPD and TetCP both are consumed at about 0.5 h.

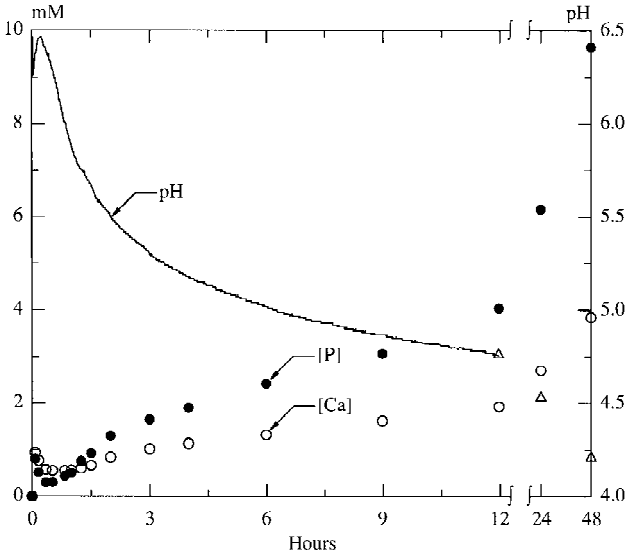


Figure 10. Variations in calcium concentration [Ca], phosphate concentration [P], and pH with time during the formation of CDHAp at 50.0°C. Calcium (open circles) and phosphate (closed circles) concentrations are plotted on the y_1 axis in m*M* (10^{-3} moles/liter). The pH (solid line to 12 h and then open triangles) values are plotted on the y_2 axis.

Comparison of stoichiometric and calcium-deficient hydroxyapatite formation

As the reaction temperature increases, two trends occur consistently for both compositions: the duration and the pH value of any given stage of the reaction decrease (Fig. 1). A decrease in the duration of each stage with increasing temperature is in accord with the large activation energies exhibited by these reactions.[9] The decrease in pH with temperature has been observed previously when stoichiometric hydroxyapatite is equilibrated with water.[12] This can be explained in terms of changes in solubilities of phases in equilibrium at the invariant point between SHAp and $Ca(OH)_2$. While calcium hydroxide is not present in these samples, SHAp will tend toward the equilibrium solution conditions that exist at this invariant point.[3] Both SHAp and $Ca(OH)_2$ exhibit retrograde solubilities, but the change with temperature is much greater for $Ca(OH)_2$[13] than for SHAp[12]. Thus, in accord with data obtained in this study, an increase in temperature will drive the equilibrium pH and equilibrium ion concentrations of SHAp to lower values. This is shown schematically in Figure 11. The invariant point labeled Pt. 1 in Figure 11 is a schematic representation of equilibrium between $Ca(OH)_2$ and SHAp at 37.4°C while Pt. 2 represents this equilibrium at 25.0°C. As shown, both compounds exhibit retrograde solubility. Both

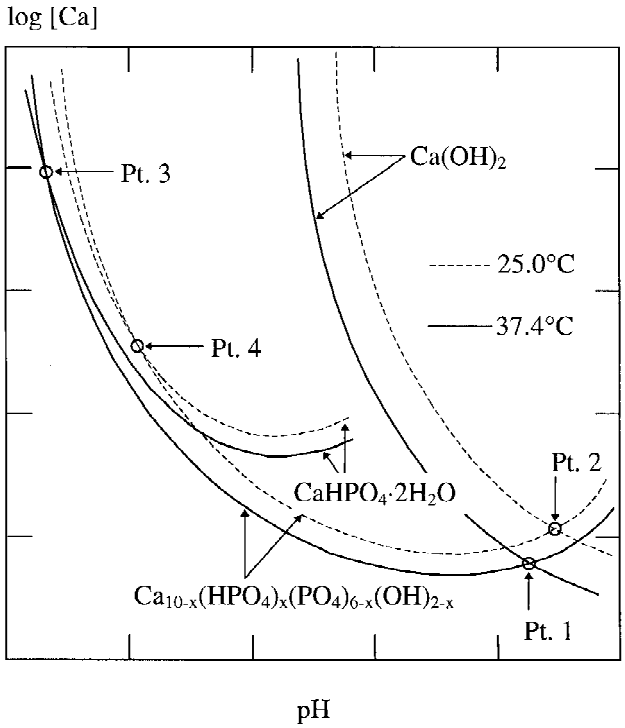


Figure 11. Schematic solubility isotherms of selected calcium phosphates showing the effects of temperature and hydroxyapatite composition on the invariant points. Dashed lines represent 25°C and solid lines represent 37.4°C.

the equilibrium pH and calcium concentrations decrease with increasing temperature.

Our present results also indicate the pH at the invariant point between CDHAp and DCPD (and DCP) decreases with increasing temperature. This is shown schematically in Figure 11 by comparing the invariant points for DCPD and CDHAp, labeled Pt. 3 (37.4°C) and Pt. 4 (25.0°C). The decrease in pH at which these two phases are in equilibrium and the increase in ion concentrations with increasing temperature is consistent with their retrograde solubilities. These results are in agreement with previous work[14] showing calculated solubility diagrams in the calcium phosphate system. An early DCPD solubility study[15] suggested the invariant point between DCPD and HAp at 37.5°C to be between 5.13 and 5.28, placing the intersection of the two respective solubility isotherms at a pH greater than that observed at 25.0°C. Later work[12,16–19] indicated that the solubility isotherm of DCPD (and DCP) intersects that of HAp at lower pH values as the temperature is increased. Although based on HAp having a fixed composition of $Ca_5(PO_4)_3OH$, the variations of the invariant point with temperature are in agreement with those observed in the present work. However, assumption that HAp is of a fixed composition is not correct. Many previous studies[6,7,20–23] have formed nonstoichiometic hydroxyapatite by a variety of low-temperature techniques, indicating that kinetically it can form. The present study has shown that CDHAp with a Ca/P molar ratio of 1.50 also is a thermodynamically stable phase because reaction, according to Eq. 1, does not produce a biphasic mixture of SHAp and a more acidic calcium phosphate, such as OCP, DCP, or DCPD. X-ray diffraction analysis of specimens equilibrated for up to 6 months after reaction at 25°C and 37.4°C indicate only HAp and has been supported by TEM studies.[24]

The results of this study also indicate that CDHAp exhibits incongruent solubility because the Ca/P ratio in the solution is not the same as in the solid. For congruent dissolution, the Ca/P ratio in solution would be 1.50 for the CDHAp synthesized in this study. The Ca/P ratio in solution at the end of the equilibration period is less than 0.5 for all three temperatures. At the equilibrium pH values attained (4.35–5.0) for CDHAp, the predominant phosphate species is $H_2PO_4^-$. Because the predominant charge-compensating ion is Ca^{2+}, the Ca/P ratio should be ~0.5 if there were no ion pair formations ($CaH_2PO_4^+$ and $CaHPO_4^0$). Therefore, mass balance requires that the Ca/P ratio in the HAp solid be slightly higher than 1.50. The bulk Ca/P molar ratios of the HAp in equilibrium with solution at 6 months are 1.51 (25.0°C) and 1.52 (37.4°C). Thus the degree of CDHAp incongruency also increases with temperature.

CONCLUSIONS

The mechanisms of CDHAp and SHAp formation at 25.0°, 37.4°, and 50°C were similar. However, the extents of the variations in the solution chemistry during HAp formation from DCPD and TetCP depend on the composition of HAp being formed and the temperature. Analysis of the calcium and phosphate concentrations indicated the solutions became critically supersaturated with respect to HAp (and NCP) at near neutral pH. Eventually the pH decreased as DCPD was preferentially consumed. Shortly before DCPD consumption was complete the pH increased as TetCP dissolution increasingly controlled the solution chemistry. X-ray diffraction analyses indicated the presence of a noncrystalline calcium phosphate and HAp at these times. Thus the first mechanistic reaction step is the complete consumption of DCPD, the partial consumption of TetCP, and the formation of HAp and NCP. The formation of the noncrystalline material was more extensive when CDHAp was the final product.

A second mechanistic reaction step involves reaction between the remaining TetCP and the two intermediate phases, NCP and HAp, to form the final HAp compositions. Continuous increases in pH were observed during this step. Regardless of the temperature or stoichiometry, hydrolysis of TetCP was rate-limiting. When the consumption of TetCP was complete, the pH slowly decreased over extended periods of equilibration. The decrease in pH for the SHAp compositions was due in part to the incorporation of carbonate while the decrease in pH for the CDHAp composition was a result of incongruency. The pH exhibited a continuous decrease over an extended period of equilibration after complete reaction to CDHAp.

The calcium and phosphate concentrations never exceeded 1.5 m*M* during HAp formation, and their variations were correlated with the disappearance of the reactants, the presence of NCP, and the formation of the HAp products. The concentrations of calcium and phosphate indicate CDHAp to be more soluble than SHAp.

This study shows SHAp and CDHAp formation *in vivo* should occur within clinically relevant periods of time. Although the SHAp composition reaches relatively high pH values upon complete reaction, the extremely low ionic strengths in solution combined with the buffering systems of the body should allow for *in vivo* formation without damage to adjacent tissues. Present results also indicate CDHAp should resorb more rapidly than SHAp *in vivo* provided surface areas and porosities of the two apatites are comparable. Thus the composition of HAp formed by cementitious reactions should be able to be used to control its *in vivo* resorption characteristics. Preliminary studies in rabbits and rats support this hypothesis.

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